

Conditioning as Group Action: Exact Compositional Conditioning at FiLM Cost

Richard Bao
richardbao419@gmail.com

Abstract

Practitioners condition networks by scaling features (FiLM) or by generating weights (hypernetworks). Both learn what each condition does. Neither learns what two conditions do together unless training data shows them together. We condition by rotating features, with rotation angles a linear function of the condition. Rotations add, so combined conditions compose automatically, and the network handles combinations it never saw. **CGA** (Conditioning as Group Action) formalizes this: the condition enters the operator linearly in an abelian Lie algebra, $T(c) = P \exp(A(Wc))P^\top$, so $T(c_1+c_2) = T(c_1)T(c_2)$ holds for any weights. A diagonal algebra yields a compositional FiLM variant; the rotation algebra with scalar position is RoPE. Under a pre-registered protocol (margins, tests, and splits committed before every run; conditioning cost within $1.2\times$ FiLM), CGA posts the lowest error on never-trained condition combinations across five transformation suites, with complete seed separation ($p=9.1\times 10^{-5}$, $N=10$) and zero-shot extrapolation to combinations longer than any combination seen in training. Swapping its linear map for an MLP raises error by up to $4.3\times 10^4\times$; the most expressive baselines generalize worst. Three registered gates returned negatives that bound the method: a fit penalty on rich scenes, a content-conditioning failure (rotations move information and cannot add it), multi-step rollouts, where every conditioner compounds error equally unless a latent-consistency loss is added, and a frozen representation, where the same task CGA wins by $8\times$ when the representation is co-trained is lost by $1.6\times$ when it is not. The negatives specify the successor. **Complex FiLM**, magnitude for content and phase for composition, beats FiLM by 36% on unseen combinations and matches it on content ($0.97\times$) under two further pre-registered gates, and on a 2D Navier-Stokes surrogate conditioned on physical parameters it passes every criterion at once: 64.6% below the best unstructured arm, every seed beating every seed, while fitting *better* in distribution ($0.938\times$) at $0.84\times$ FiLM's cost. A further gate bounds that: the fit penalty tracks how much content the condition must supply, not the operator's geometry, and no variant we tested repairs it when the condition supplies content itself. The same structure yields a guidance knob needing no second forward pass, which distorts $2\text{--}5\times$ less than classifier-free guidance on a deterministic target, though not on a multimodal one. Sweeping the number of simultaneously changed conditions and their magnitude independently isolates the mechanism: the advantage is $+0.6\%$ on one condition and rises to $+46.6\%$ on four, and is absent at weak single conditions, so it is composition rather than difficulty or strength. It is also narrower than we expected. Both screening quantities passed on a sixteen-control chain built because the account favoured it, and the advantage was flat at -1.1% : structured conditioning pays when a task's own conditions compose the way the operator does, which no dimensional criterion captures.

1 Introduction

Nearly every conditional model must inject something into its computation: a class, an action, a timestep, a text embedding. Call this conditioning. Diffusion models modulate every block on the timestep and class via adaLN [2]; visual reasoning, speech, and RL systems modulate features on questions, speakers, and goals [1]; world models condition latent dynamics on actions. Three mechanisms carry almost all of this conditioning: concatenation, feature-wise affine modulation (FiLM), or a hypernetwork that emits weights from the condition.

Two of these bracket an expressivity axis; we set concatenation aside, and carry it as a baseline arm. FiLM's diagonal transform is cheap and stable but cannot mix features. Hypernetworks can express any linear operator but pay quadratic cost and, as we show, memorize the training conditions: their error on unseen combinations of familiar conditions is worse than FiLM's, despite $43\times$ the conditioning compute. The middle of this axis, structured operator families between diagonal and dense, is well explored for weight fine-tuning [6–9] and largely unexplored in FiLM's role: per-sample, input-conditioned modulation of activations.

Conditioning plays two roles, and the distinction runs through this paper. A condition can name a *change* to apply, such as a pose offset or an edit, or it can name the *content* to produce, such as a class or a caption. FiLM serves both.

We ask which structure, at FiLM-level cost, buys compositional generalization: correct behavior on combinations of conditions never seen together in training. Our answer imports a mechanism from positional encoding. RoPE [4] and its generalization GRAPE [5] encode position n as a group action $G(n) = \exp(n\omega L)$; because position enters linearly in the Lie algebra, $G(n+m) = G(n)G(m)$ holds exactly. We transplant this construction from scalar positions in attention to arbitrary condition vectors in activation modulation.

Contributions. (1) **Framework.** CGA treats conditioning as a group representation of the condition space; FiLM, RoPE, and GRAPE drop out as special cases, and a two-line proposition makes condition composition exact for any weights (§3). (2) **Protocol.** A pre-registered, budget-fair evaluation: margins, statistical tests, and single-read splits committed to version control before every run, with a shared counter enforcing parameter and FLOP parity; we report every verdict, including negatives against our own predictions (§4). (3) **Where it works, and where it stops.** CGA has the lowest error on unseen condition combinations in all five suites where the condition is a transformation, by up to five orders of magnitude, and in every image suite the more expressive the conditioner, the worse it generalizes. Three of the eight suites returned pre-registered negatives, and each names a setting where the composition law of Proposition 1 buys nothing: 3D Shapes, which CGA wins on unseen combinations but loses on training fit; content conditioning; and multi-step rollouts (§5). (4) **Successor and guidance.** The negatives specify Complex FiLM, validated under two further gates as a FiLM replacement covering both conditioning roles, whose phase channel also yields guidance as an exact operator power: 2–5× less distortion than CFG with no unconditional pass (§5). (5) **A predictive account of mechanism choice.** One rule generates the wins and losses. Approximation error is class membership rather than capacity, and admits closed forms per mechanism through the polar decomposition; exact composition bounds extrapolation error by $\|A\| \|\hat{W} - W\| \|c\|$, independent of how far the condition sits from those seen, and only on the compact factor. Both quantities are estimable from short runs. They are not sufficient: the first prediction we made before seeing an outcome failed, and identified a third condition we could already measure: whether the task’s own conditions compose the way the operator does (§6).

2 Related work

FiLM and conditional normalization. FiLM [1] and its normalized variants (conditional BN/LN, adaLN in DiT [2]) apply per-feature affine transforms generated from a condition. All are diagonal. We keep FiLM’s role and enrich only the operator.

Structured operators in parameter-efficient fine-tuning (PEFT). Low-rank, orthogonal, and butterfly operator families are well studied as weight fine-tuning: LoRA [6], OFT [7], BOFT [8], HRA [9], Group-and-Shuffle [10]; He et al. [11] give a unified view. These learn one fixed transform per task. Our setting differs on every axis that matters: per-sample, input-conditioned, activation-space, and evaluated for compositional generalization. Hypernetworks [3] generate weights from conditions and inherit the memorization behavior we quantify.

Position encodings as group actions. RoPE [4] rotates query-key pairs by angles linear in position; GRAPE [5] generalizes this to group representations with exact relative laws. The mechanism is theirs; our contribution is the transplant to arbitrary condition vectors in FiLM’s role, with a learned basis, and the budget-fair compositional evidence.

Compositional generation. Montero et al. [12] showed reconstruction-trained generative models fail on unseen factor combinations. Score and energy composition [23] composes concepts across separate conditional passes; we compose conditions inside the operator, in one pass. Symmetry-based representation learning [13, 14] argues latents should carry group actions; we supply the conditioning-side mechanism.

Nearest prior work. We searched deliberately for work that would invalidate this paper. The closest hits: Feature-space rotations with exact composition exist for geometric transformations [17]; abelian exponential-map operators for disentanglement [22]; attribute operators for recognition [26]; multilinear attribute mixing for unseen combinations [27]; per-sample amplitude-and-phase activation modulation in periodic implicit networks [20]; and composition by complex phase addition in knowledge-graph embeddings [21]. Rotation-structured latent world models with latent-prediction losses originate with Quessard et al. [15], Caselles-Dupré et al. [16], and the homomorphism autoencoder

[18], whose composition property is emergent rather than imposed; latent consistency objectives are standard in model-based RL [19]. Distillation moves guidance into the conditioning path [25], and prior work documents CFG’s high-scale distortion [24]. We claim none of these components. We claim the combination: a group-structured operator in FiLM’s role at budget parity, composition imposed exactly by a linear-in-algebra map, magnitude split from phase as content and composition channels, guidance as an exact power, and verdicts from pre-registered gates.

3 The CGA framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM computes $y = \gamma(c) \odot x + \beta(c)$; hypernetworks emit $W(c)$. All are instances of

$$y = T(c)x + \beta(c), \quad (1)$$

and they differ only in where the per-sample operator $T(c)$ comes from.

Definition 1 (Compositional conditioner). $T : \mathbb{R}^k \rightarrow \text{GL}(d)$ is compositional if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1 + c_2) = T(c_1)T(c_2)$ and $T(0) = I$.

Proposition 1 (Exact composition by construction). Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $A : \mathbb{R}^m \rightarrow \mathfrak{a}$ linear, $W \in \mathbb{R}^{m \times k}$, and P orthogonal. Then $T(c) = P \exp(A(Wc)) P^\top$ is a compositional conditioner for every W, P . If $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.

Proof. $\exp(X)\exp(Y) = \exp(X+Y)$ for commuting X, Y ; $A \circ W$ is linear; conjugation preserves products; $\exp(0) = I$. $\square$

Composition is usually a behavior the network must learn from data. Here it is a property of the parametrization. It holds at initialization, during training, and far outside the training distribution. Training only decides which directions the condition rotates and how fast.

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms:

$\mathfrak{a}$	condition	P	recovered mechanism
$\text{diag}(\mathbb{R}^d)$	c	I	<i>exponential FiLM</i> : $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	this work (FiLM’s role)
$\text{diag} \times \mathfrak{so}(2)^{d/2}$	general c	canonical	Complex FiLM (§5)

Standard FiLM reaches the same diagonal operators through an MLP head $\gamma(c)$. The operators match; only the composition law is missing, and §5 shows that its absence is what breaks FiLM on unseen combinations.

Cost. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, $\exp$ is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis P is the whole budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is $1.52\times$ FiLM’s entire conditioning path at $d=128$. Two fixes work. A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to $1.11\times$ FiLM, with 4,800 parameters against the 8,128 dimensions of $\text{SO}(128)$; that is enough, because P only needs the planes the condition acts on. Inside a transformer, no explicit P is needed: the adjacent dense projections can absorb P , exactly as RoPE relies on the query and key projections rather than on a basis of its own. Linearity also gives the operator a strength dial for free: $T(c)^w = T(wc)$, so raising the operator to a power is the same as scaling the condition, and §5 uses this in place of CFG’s extrapolation between a conditional and an unconditional prediction. The operator equals the identity at initialization and at $c=0$, and every singular value is 1. Unit tests pin all three properties. Complex FiLM extends the algebra to $\text{diag} \times \mathfrak{so}(2)^{d/2}$: each feature pair is multiplied by $m e^{i\theta}$, with magnitude from FiLM’s expressive head (the content channel) and phase linear in the condition (the composition channel), at $0.84\times$ FiLM’s cost in our harnesses.

4 Pre-registered, budget-fair protocol

Vocabulary. An *arm* is one conditioning method, trained under the shared recipe. A *suite* is one experiment: a fixed backbone, dataset, and training loop that all of its arms share. A *gate* is that suite’s pass-or-fail rule, written down before the run as a conjunction of *criteria*: typically a margin over the strongest baseline on unseen combinations, a no-regression criterion on the training distribution, and a budget criterion. A suite passes only if all of its criteria pass, and we report the outcome either way.

Design. Every suite is pre-registered: success margins, statistical tests ($N=10$ seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff’s $|\delta| \geq 0.474$), and data we fixed and committed margins, tests and splits to version control before each run. Each test split is read once, after all selection on a validation split. All arms share one backbone and one training recipe. A shared counter enforces the budget: our conditioning path stays within $1.20\times$ FiLM’s FLOPs and $1.05\times$ the smallest hypernetwork baseline’s parameters; baselines may be larger, and reach $48\times$ our parameter count. We record deviations as dated amendments in the committed specifications, plus one accounting erratum that our own audit caught and that downgraded a favorable verdict.

Threats to validity, and the design answer. The benchmarks are self-constructed, so we built them to remove judgment from scoring: dSprites and 3D Shapes are deterministic factor-to-image maps, so every held-out condition has a unique ground-truth image and pixel error is an objective compositional metric, with no perceptual score and no human rating. Pre-registration removes outcome-contingent analysis; the verdicts below include three negatives, one of which falsified our own headline prediction. Every number in this paper regenerates from the committed logs with one command per suite; re-running a suite from scratch costs one training run per arm per seed.

Arms. FiLM; conditional LayerNorm; Concat-MLP; a dense hypernetwork $I+\Delta W(c)$; a dynamic low-rank map $I+A(c)B(c)^T$; the Lie operator; Complex FiLM (both variants); and an MLP-head ablation of our own method (same basis, same operator family, more compute, no linearity).

The experiments. **Two synthetic suites.** Learn $y = M(c)x$, where c picks a combination of 8 rotation primitives. The second hides the primitives behind a random basis, so the operator has to find them, and additionally tests all 56 never-trained triples. **Three image suites** on dSprites and 3D Shapes. Encode a real image, apply $T(\Delta)$ to the learned latent for a factor change Δ , decode, and score pixel error against the true transformed image; OOD means never-trained *types* of two-factor change. Two of the three add a categorical shape factor. **Content.** A small diffusion transformer generates dSprites images from a full factor specification, so the condition names what to draw rather than how to change it. **Rollouts.** A world model applies the conditioning arm recurrently, one action per step, tested at horizons 10 and 20 after training on 3. **The successor.** Complex FiLM, through the dSprites and content setups. **Guidance.** Condition-dropout training, then classifier-free guidance at strength w for FiLM against condition powering for Complex FiLM, scored against deterministic ground truth at each strength.

5 Results

(1) Composition is exact, and the weights say what each condition does. On the hidden-basis suite, CGA reaches 3.3×10^{-8} error on unseen pairs and 5.3×10^{-8} on never-trained triples. The best baseline sits five orders of magnitude higher, and every baseline degrades as combinations lengthen. Learned angles match the true generators to 2×10^{-4} rad, so reading the weights tells you which plane each condition rotates and how fast (Figure 4).

(2) The advantage comes from composing conditions, and its shape shows it. Comparing arms on a fixed evaluation cannot separate a compositional mechanism from an arm that is simply better. We therefore swept the number of simultaneously changed factors and their magnitude independently, and registered in advance that the advantage must vanish for a single weak condition and grow with the number composed. At the trained magnitude the operator gains 0.6% over the best additive arm on one factor, then 25.5%, 38.7% and 46.6% on two, three and four ($p=9.1\times 10^{-5}$, $\delta=-1.00$); at weak single conditions every arm falls within 1.6% of the others. An advantage that is absent at one condition and rises with the number composed cannot be difficulty, strength, or a generally stronger arm.

Table 1: Error on never-trained condition combinations: mean (seed sd) over 10 seeds. CGA is lowest in every column, with complete seed separation against the strongest hypernetwork baseline ($p=9.1\times 10^{-5}$, $\delta=-1.0$). OOD means out-of-distribution: a condition combination never seen in training. Every column except 3D Shapes passed its full pre-registered gate; 3D Shapes failed on a separate training-fit criterion (text). The last column reruns the dSprites setup with Complex FiLM. **Columns are not comparable to one another**: the two synthetic suites report operator MSE, the rest report per-pixel MSE.

Conditioner	Aligned	Hidden basis	dSprites	+shape	3D Shapes	Complex FiLM
FiLM	$1.3 (0.1)\times 10^{-2}$	$1.6 (0.04)\times 10^{-2}$	$3.0 (0.2)\times 10^{-3}$	$5.9 (0.2)\times 10^{-3}$	$4.5 (1.0)\times 10^{-3}$	$3.1 (0.2)\times 10^{-3}$
Concat-MLP	$1.5 (0.3)\times 10^{-2}$	$1.1 (0.4)\times 10^{-2}$	$2.9 (0.1)\times 10^{-3}$	$6.3 (0.2)\times 10^{-3}$	$5.4 (5.2)\times 10^{-3}$	–
Cond-LN	$2.4 (0.1)\times 10^{-2}$	$2.7 (0.05)\times 10^{-2}$	$3.2 (0.2)\times 10^{-3}$	$6.1 (0.3)\times 10^{-3}$	$8.6 (13.1)\times 10^{-3}$	–
Hypernetwork	$1.9 (0.3)\times 10^{-3}$	$3.4 (0.9)\times 10^{-3}$	$3.8 (0.3)\times 10^{-3}$	$7.0 (0.4)\times 10^{-3}$	$9.3 (2.3)\times 10^{-3}$	$3.8 (0.2)\times 10^{-3}$
Dyn. low-rank	$5.7 (0.5)\times 10^{-3}$	$6.1 (0.4)\times 10^{-3}$	$5.1 (0.9)\times 10^{-3}$	$8.1 (0.4)\times 10^{-3}$	$2.0 (0.4)\times 10^{-2}$	–
MLP-head (abl.)	–	$1.1 (0.2)\times 10^{-3}$	$2.7 (0.2)\times 10^{-3}$	$5.0 (0.2)\times 10^{-3}$	$3.3 (0.8)\times 10^{-3}$	–
Complex FiLM (linear mag.)	–	–	–	–	–	$2.0 (0.1)\times 10^{-3}$
Complex FiLM	–	–	–	–	–	$2.0 (0.2)\times 10^{-3}$
CGA (ours)	$7.4 (2.0)\times 10^{-4}$	$3.3 (2.2)\times 10^{-8}$	$1.8 (0.1)\times 10^{-3}$	$4.0 (0.1)\times 10^{-3}$	$1.2 (0.2)\times 10^{-3}$	$1.8 (0.1)\times 10^{-3}$

Every conditioner degrades on unseen condition combinations. CGA degrades least.

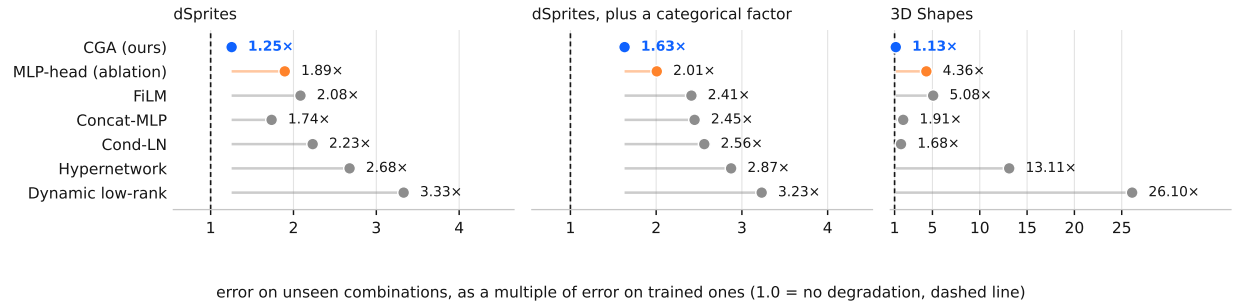


Figure 1: How much worse each conditioner gets when conditions are combined in ways it never saw, as a multiple of its own error on trained combinations. A value of 1.0, the dashed line, would mean no degradation at all. CGA sits closest to it on all three image suites, and the two most expressive conditioners sit furthest: their capacity goes into memorising the training combinations rather than generalising past them.

The magnitudes follow. On dSprites every arm fits the training distribution equally well, which isolates composition: CGA cuts error on unseen combinations by 53.8% against the strongest baseline at 39× fewer conditioning FLOPs, and by 42.9% once a categorical shape factor joins the condition. Read as a multiplier, going from trained to unseen combinations costs CGA 1.25×, or 1.63× with the shape factor, against 2.1–3.3× for every baseline. We call that multiplier the *compositional gap*.

(3) The linear map is the mechanism, and capacity works against it. The ablation keeps the basis and the operator family, spends more compute, and swaps only the linear map for an MLP. It loses by $4.3\times 10^4\times$ on synthetic triples and 1.55× on dSprites. The two hypernetwork arms can express any operator and post the largest compositional gaps of any arm on every image suite, 2.7–3.3× on dSprites and 13–26× on 3D Shapes against CGA’s 1.13–1.63×. Their capacity memorises the training combinations.

(4) Complex FiLM covers both conditioning roles at lower cost. Complex FiLM multiplies feature pairs by $m e^{i\theta}$, giving content to the magnitude channel and composition to the phase. Under two pre-registered gates it beats FiLM by 36% on unseen change types at fit parity and 0.84× FiLM’s cost, and matches FiLM on content generation at 0.97× its error (pre-registered non-inferiority margin 1.10×), where the pure rotation failed by 7×. Giving the magnitude

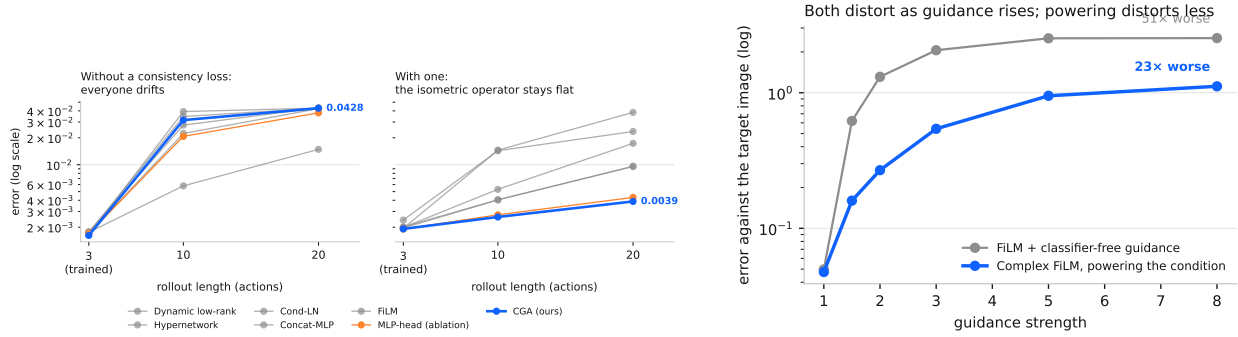


Figure 2: **Left:** error against rollout length, trained only to length 3. Without a consistency loss every conditioner drifts at much the same rate. Adding one separates them: the isometric operator flattens out while the baselines keep climbing. **Right:** what turning guidance up does to the generated image. Both distort, since the target here is deterministic and there is no distribution to sharpen, but classifier-free guidance distorts twice as fast.

channel a linear map instead of an MLP fails that same content metric (0.19 against FiLM’s 0.04), so content needs the expressive head. FiLM is the magnitude channel on its own; the full complex multiplication does both jobs.

A third gate confirms it outside the disentanglement benchmarks. On a 2D Navier-Stokes surrogate conditioned on four physical parameters (viscosity, drag, forcing and background advection, with the initial vorticity field supplying all of the content), Complex FiLM passes every criterion on fresh seeds: 45.4% below FiLM, 64.6% below the best unstructured arm at $\delta=-1.00$, and an in-distribution fit ratio of **0.938** $\times$, the first arm here to fit better than the unstructured baseline. The margin replicates across independent seed sets (65.3% and 1.030 $\times$ on the first, 64.6% and 0.938 $\times$ on the second). The claim covers Complex FiLM and not the pure group operator: its magnitude head is nonlinear in c , so composition stays exact on the phase channel only. It buys fit by spending half the guarantee, and the arm that keeps the guarantee on both channels failed this criterion twice.

(5) Content is provably out of reach for an orthogonal conditioner. In the content suite, rotation conditioning fails: pixel MSE 0.31 against FiLM’s 0.04, an 8 $\times$ regression, and the ablation fails identically. A proof explains it. Producing content means suppressing features, so the required transformation carries eigenvalues of modulus far from 1; every orthogonal matrix has all $|\lambda|=1$; and a change of representation acts by conjugation, which preserves the spectrum. No orthogonal conditioner can carry “generate this image” in *any* representation, and co-training cannot repair it (§6).

(6) The remaining fit penalty tracks content demand. Where the condition supplies some content, the operator fits worse in distribution than an unstructured arm. On 3D Shapes CGA posts the lowest error on unseen combinations (87.5% below the hypernetwork) and still trips the pre-registered no-regression criterion at 1.46 $\times$. On the physics task the same criterion fails at 1.118 $\times$. We registered two relaxations against it that keep exact composition, adding a scaling generator and conjugating the orthogonal basis by a learned diagonal. The conjugated operator looked like a cure on validation (0.962 $\times$) and reached 1.1007 $\times$ on the held-out split over ten seeds, missing by 0.065 percentage points. Relaxing the metric does not fix it.

A magnitude channel does, but only where the condition supplies no content. Complex FiLM reaches 0.938 $\times$ on the physics task, whose condition is four parameters acting on content the input already carries; gated on fresh seeds on 3D Shapes, whose condition includes a categorical shape swap, it reaches 1.542 $\times$, slightly worse than the pure rotation, while still passing the compositional criteria by 84.5% at $\delta=-1.00$. How much content the conditioning must supply governs the penalty. This family of operators suits conditions that *transform* content supplied elsewhere, and no variant we tested rescues it otherwise.

(7) The advantage requires a co-trained representation. We ran the same dataset, held-out attribute pairs, arms and budget counter twice, varying whether the representation trains alongside the operator. We previously called this a single-variable comparison and it is not: the co-trained run optimises a pixel objective and scores in pixel space, while the frozen run optimises and scores in latent space, so representation, objective and metric all differ. The within-run ratios are the defensible part. Co-trained, CGA’s error on unseen combinations sits 87.5% below the hypernetwork’s.

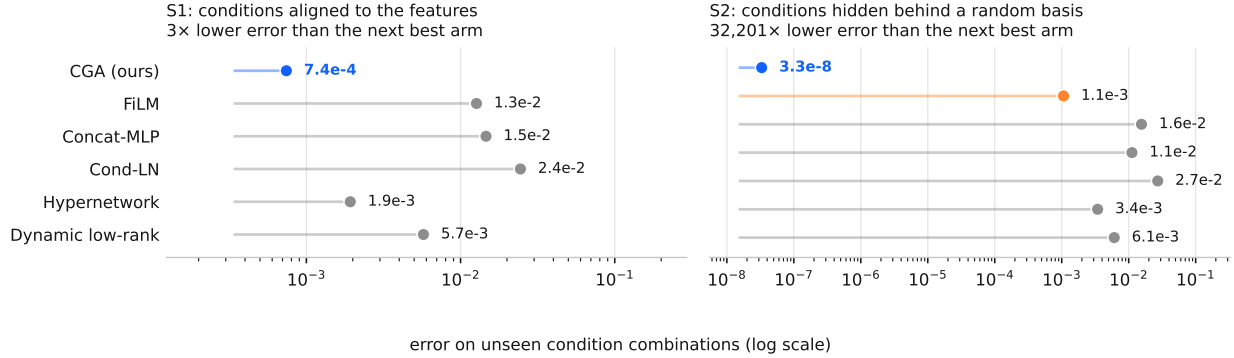


Figure 3: On the synthetic suites, where the conditions form a group exactly, the gap is not a matter of percentages. Each dot is one conditioner’s error on unseen combinations; note the log axis. Hiding the conditions behind a random basis (right) costs CGA nothing, because it learns the basis, while the gap to the next best arm grows to five orders of magnitude.

Frozen, against a plain autoencoder trained once for reconstruction and then shared unchanged by every arm, CGA lands 60.1% worse than that hypernetwork while using 48× fewer conditioning parameters, and fits 2.43× worse in distribution against a 1.10× ceiling. Complex FiLM fails there too, so the rotation channel is not the culprit.

The operator’s advantage is therefore a property of the operator and its representation together. Bolted onto a frozen general-purpose encoder, which is the pattern latent-space editing deploys, it loses. CGA still posts the smallest compositional gap of any arm that fits (1.39× against the hypernetwork’s 2.11×): it degrades most gracefully and still loses, because grace does not recover a fit handicap.

(8) Rollouts need a consistency loss, and the operator alone does nothing. In the rollout suite every arm compounds error at the same rate, because each step adds a small mismatch between the predicted and the true latent. A contracting operator shrinks past mismatches; an isometry preserves them, so they accumulate. Training with $\lambda \|T(a)z_s - z_{s+1}\|^2$, applied to every arm equally, inverts the picture: the rotation operator’s horizon-20 error goes flat at 0.0039 (growth 2.0×) against the hypernetwork’s 0.038 (growth 19.8×) and FiLM’s 0.017 (Figure 2, left). The loss is standard in model-based RL [19], and rotation transitions with latent-prediction objectives appear in the homomorphism autoencoder [18]. We separate the two effects: adding the same loss to every arm at the same budget shows that neither the loss nor the isometric operator fixes rollouts alone, and that they work only together. A null result sharpens this. Shrinking the latent by a fixed factor at every step, swept from 0.003 to 0.1, leaves horizon-20 error unchanged, so whatever removes the accumulated mismatch must depend on the state. A separate single-step criterion fails, so the rollout suite counts as a negative in Table 1 even though its horizon-20 result is favorable.

(9) Guidance costs one forward pass, and holds only on a deterministic target. CFG’s distortion at high scales is documented [24]; our benchmark isolates it against an exact alternative, because the conditioning target is deterministic. CFG’s error grows 50.7× from strength 1 to 8, while powering the condition through the phase channel grows 23.7× ($p=1.6 \times 10^{-4}$), stays 2–5× lower at every strength above 1, matches at native strength (0.96×), and needs no unconditional pass and no distillation [25] (Figure 2, right). Neither method improves over strength 1 here: the target is deterministic, so no distribution exists to sharpen, exactly as the suite’s registered caveat stated.

The sweep in (2) refutes the other half of what we registered. We had argued that additive aggregation, which is what a value-sum over condition tokens performs, should degrade as $\frac{1}{2} \|\sum_i A_i\|^2$ and therefore worsen with conditioning *strength*. The advantage instead reverses beyond the trained scale, reaching −14.8%. Rotations are periodic, so past the trained angle the operator wraps and the error oscillates. The bound of §6 survives as an upper bound linear in $\|c\|$, but it is loose exactly where we relied on it. Exact composition buys accuracy when stacking conditions at the scale trained on, and gives no strength dial.

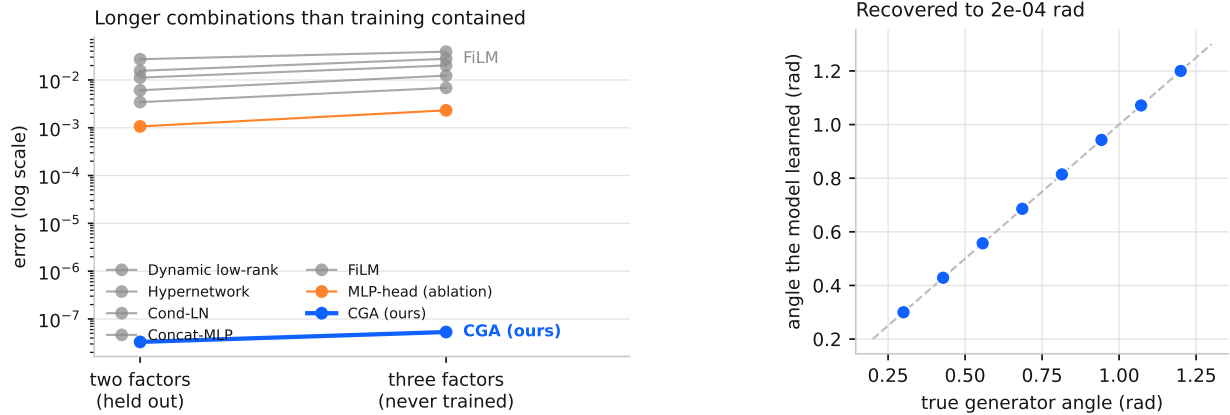


Figure 4: **Left:** what happens when a test combination is longer than anything in training. Every baseline gets worse; CGA does not move, because composition is a property of its parametrization rather than something it had to learn. **Right:** the angles the model learned for each condition, against the true generators of the data. Reading the weights tells you which plane a condition rotates, and how fast.

6 Why: a predictive account of mechanism choice

The results above are a table of wins and losses. This section gives the rule that generates them, and it is the part we expect to outlast the operator. Full derivations are in the released `THEORY.md`.

Write a conditioning mechanism as a family of feature maps $\{T_c\}$ and split test error as $E_{\text{approx}} + E_{\text{est}} + E_{\text{extrap}}$. Two facts then determine which mechanism suits which task.

Approximation error is class membership, not capacity. The reachable sets are nearly disjoint rather than nested: the diagonal maps and the orthogonal maps intersect only in the 2^d matrices with entries ± 1 . Switching from FiLM to a rotation does not shrink the reachable set, it *moves* it, so parameter counts are the wrong currency. Complex FiLM uses $0.85\times$ FiLM’s parameters while substituting one operation class for another. The right coordinates are polar, $T = US$, and each mechanism reaches one factor. This yields exact minimisers, not bounds: $E_{\text{approx}} = 0$ for a hypernetwork, $\|T^* - \text{diag } T^*\|_F$ for FiLM, $\|S^* - I\|_F$ for a rotation (orthogonal Procrustes), and a blockwise projection residual for Complex FiLM. So a rotation’s approximation error *is* the stretch content of the task’s operator, which is why an isometry fails at content by $8\times$ at any parameter count.

Exact composition bounds extrapolation, and only on the compact factor. For $T(c) = \exp(A(Wc))$ with A skew, the exponential is 1-Lipschitz, giving $\|\hat{T}(c) - T(c)\|_F \leq \|A\| \|\hat{W} - W\| \|c\|$. The bound depends on $\|c\|$ alone and not on how far c sits from the training conditions: reaching an unvisited *combination* costs nothing beyond reaching the same magnitude, and if the training conditions span the condition space then W is identifiable and the error vanishes with data. With a symmetric part present the Lipschitz property fails and error grows like e^s , which is why the magnitude channel needs a clamp while the phase channel composes to 10^{-6} , and why guidance degrades linearly in strength on the rotation channel and exponentially on the magnitude one.

One impossibility, and its converse. If the required transformation has any eigenvalue of modulus $\neq 1$, no orthogonal conditioner can represent it in *any* representation: a change of basis is a conjugation, and conjugation preserves the spectrum. Content conditioning needs $|\lambda| \rightarrow 0$, so its $8\times$ failure is structural rather than incidental. The converse matters as much. Approximation error refers to the transformation required *in the coordinates the encoder produced*, so it is a property of the task and representation together, not of the task alone. A learned representation can repair a rotation wearing the wrong basis and can never repair a change of scale. This is why freezing the representation was fatal, and it carries a design rule: the operator must be the *only* conditioning path, since a parallel unstructured route removes the pressure to find such coordinates. Our own hybrid arm, which contained a concatenation path, scored $+1.7\%$ against concatenation alone, which is a tie.

The rule, and what it explains. Choose the mechanism whose reachable factor matches the polar type the task requires; then adopt exact composition only if the evaluation visits unvisited conditions. Short runs estimate both quantities: in-distribution fit ratio for the first, compositional gap for the second. Across our nine comparisons the two jointly separate every win from every loss, and each is independently observed to fail while the other passes: the camera suite had an acceptable fit and no gap, the audio suite a fit ratio of 1.000 and no gap, PDEBench an enormous gap and an unusable fit. The account also covers results the capacity framing could not, including why Complex FiLM specifically beats a pure rotation on a task whose operator carries both viscous decay (stretch) and advection (rotation).

The two checks are not sufficient, and the first out-of-sample test says so. Both we calibrated both thresholds above on the nine comparisons they summarise, so that agreement is consistency, not predictive accuracy. We then built the case the account favours: a sixteen-control parametric chain, the high-dimensional regime where coverage arguments say structure should matter most. Both checks passed before the run, with a compositional gap of $1.83\times$ and a fit ratio of $1.011\times$, so the account predicted a win. The measured advantage was flat at -1.1% across one, two, four and eight simultaneous controls.

A third condition is missing, and the instrument for it was already in the repository. A rigging check we had added only to confirm a task was not built to flatter the operator found 0 of 12 control pairs matching the operator’s form. That is the class test of the previous paragraph applied to the *task* rather than to the mechanism, and it called the outcome before training. The reason is visible in hindsight: dSprites factors are independent coordinates of the generative process, so combining two changes yields a well-defined image and composition holds exactly in factor space, whereas the chain’s saturation and dynamics stages interact, so the combined effect is not the composition of the individual effects. There is then nothing for exact composition to be exact about, however large the condition space or the gap. Structured conditioning pays when the task’s own conditions compose the way the operator does, which is narrower than any dimensional criterion and is measurable in advance.

Honest status. The mathematics is elementary: Procrustes, Cauchy, and a standard Lipschitz property of the exponential on compact groups. We calibrated the thresholds quoted above on the same nine tasks they summarise, so the framework is consistent with our evidence rather than validated out of sample, and the tasks themselves were not pre-registered even though the splits were. We state the rule so that it can be falsified, and give in §8 the experiment that would do it.

7 Practical guidance

The account yields a procedure, and two short runs decide it. Fit one high-capacity conditioner and read its effective operator; the closed forms of §6 then give each mechanism’s approximation error, and the polar type says which class suits the task. Measure the compositional gap on your own evaluation. Then check that the task’s own conditions compose the way the operator does, which §6 shows is the condition our first out-of-sample prediction missed.

Reach for a group-structured conditioner when all three hold: the condition names a change to apply rather than content to produce, held-out combinations are meaningfully harder than trained ones, and the representation trains alongside the operator. Keep FiLM otherwise. Complex FiLM is the drop-in where the condition transforms content the input already supplies, at lower cost than FiLM. In rollouts, pair the operator with a latent-consistency loss, since neither works alone.

8 Limitations

All evidence comes from controlled 64×64 benchmarks with known factors, width-128 models, and a single conditioning site; we validate nothing at production scale, and the natural deployment experiment is the adaLN site of a text-conditioned diffusion transformer. Learned latents do not support exact group action, so the advantage on real images is the $1.8\text{--}8\times$ error reduction quoted above, not the $10^5\times$ of the synthetic suites. 3D Shapes shows orthogonality can cost in-distribution fit on rich scenes. Conditions far from any group action, such as open-ended text, remain untested; the learned-condition-geometry extension is future work.

The guidance result does not transfer to a multimodal target. Our guidance benchmark measures distortion against a deterministic target, so it cannot see the distribution-sharpening CFG exists to provide. A second harness makes $p(x \mid c)$ uniform over 64 cells, so fidelity and diversity are separately measurable against a target known in closed form. An unregistered single-seed pilot on it, with no gate and no verdict claimed, goes against us on all three axes we could measure: at matched fidelity CFG retains more diversity, the strength knob is not monotone because the rotation channel wraps, and the base model fits 33% worse before any guidance at all. We therefore claim less distortion at equal cost on a deterministic target, and no more.

The experiment that would falsify the account. The rule of §6 predicts a *location*, not a direction. Build a task family whose conditional operator has a dialable stretch-to-rotation ratio, $T^*(c) = U(c)^\alpha S(c)^{1-\alpha}$; the closed forms then predict where FiLM, Complex FiLM and a rotation each take over as α sweeps. Registering those crossover points before running would convert a framework fitted on nine observations into a prediction tested out of sample, and would predict our own losses. We have not run it.

Reproducibility statement. We release the code, the pre-registrations, and the raw logs. Every suite has a specification committed before its decision run; a registry maps each suite label in this paper to the module that produced it, the log it wrote, and its verdict, so a replicator reproduces any row with a single command. Every number, table, and figure here regenerates from those append-only logs. Unit tests pin the operator invariants (identity at initialization and at $c = 0$, orthogonality, exact composition), the budget counters, and the registry itself. All amendments and one accounting erratum, a FLOP undercount our own audit caught that downgraded a favorable verdict, we disclose with the results.

References

- [1] E. Perez, F. Strub, H. de Vries, V. Dumoulin, A. Courville. FiLM: Visual reasoning with a general conditioning layer. *AAAI*, 2018.
- [2] W. Peebles, S. Xie. Scalable diffusion models with transformers. *ICCV*, 2023.
- [3] D. Ha, A. Dai, Q. Le. Hypernetworks. *ICLR*, 2017.
- [4] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, Y. Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv:2104.09864*, 2021.
- [5] L. Zhang, K. Chen, X. Liu, et al. Group representational position encoding. *ICLR*, 2026.
- [6] E. Hu et al. LoRA: Low-rank adaptation of large language models. *ICLR*, 2022.
- [7] Z. Qiu et al. Controlling text-to-image diffusion by orthogonal finetuning. *NeurIPS*, 2023.
- [8] W. Liu et al. Parameter-efficient orthogonal finetuning via butterfly factorization. *ICLR*, 2024.
- [9] S. Yuan, H. Liu, H. Xu. Householder reflection adaptation. *NeurIPS*, 2024.
- [10] M. Gorbunov et al. Group and shuffle: Efficient structured orthogonal parametrization. *NeurIPS*, 2024.
- [11] J. He, C. Zhou, X. Ma, T. Berg-Kirkpatrick, G. Neubig. Towards a unified view of parameter-efficient transfer learning. *ICLR*, 2022.
- [12] M. Montero, C. Ludwig, R. Costa, G. Malhotra, J. Bowers. The role of disentanglement in generalisation. *ICLR*, 2021.
- [13] I. Higgins et al. Towards a definition of disentangled representations. *arXiv:1812.02230*, 2018.
- [14] T. Cohen, M. Welling. Group equivariant convolutional networks. *ICML*, 2016.
- [15] R. Quessard, T. Barrett, W. Clements. Learning disentangled representations and group structure of dynamical environments. *NeurIPS*, 2020.

- [16] H. Caselles-Dupré, M. Garcia-Ortiz, D. Filliat. Symmetry-based disentangled representation learning requires interaction with environments. *NeurIPS*, 2019.
- [17] D. Worrall, S. Garbin, D. Turmukhambetov, G. Brostow. Interpretable transformations with encoder-decoder networks. *ICCV*, 2017.
- [18] H. Keurti, A. Neitz, S. Weichwald, B. Schölkopf, et al. Homomorphism autoencoder: Learning group structured representations from observed transitions. *ICML*, 2023.
- [19] N. Hansen, X. Wang, H. Su. Temporal difference learning for model predictive control. *ICML*, 2022.
- [20] E. Chan, M. Monteiro, P. Kellnhofer, J. Wu, G. Wetzstein. pi-GAN: Periodic implicit generative adversarial networks. *CVPR*, 2021.
- [21] Z. Sun, Z. Deng, J. Nie, J. Tang. RotatE: Knowledge graph embedding by relational rotation in complex space. *ICLR*, 2019.
- [22] X. Zhu, C. Xu, D. Tao. Commutative Lie group VAE for disentanglement learning. *ICML*, 2021.
- [23] N. Liu, S. Li, Y. Du, A. Torralba, J. Tenenbaum. Compositional visual generation with composable diffusion models. *ECCV*, 2022.
- [24] A. Bradley, P. Nakkiran. Classifier-free guidance is a predictor-corrector. *arXiv:2408.09000*, 2024.
- [25] M. Fu et al. TeEFusion: Blending text embeddings to distill classifier-free guidance. *ICCV*, 2025.
- [26] T. Nagarajan, K. Grauman. Attributes as operators. *ECCV*, 2018.
- [27] M. Georgopoulos, G. Chrysos, M. Pantic, Y. Panagakis. Multilinear latent conditioning for generating unseen attribute combinations. *ICML*, 2020.